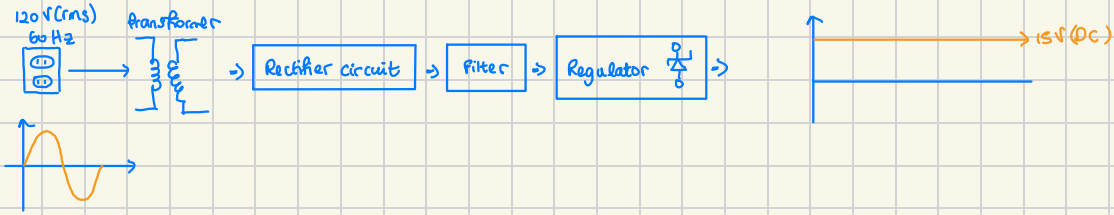
ELG 2136

Lecture 5

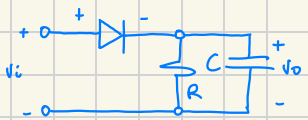
Rectifiers

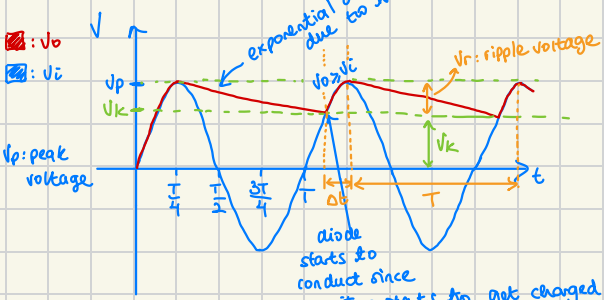
Rectification process



Filter Component [Ideal diode]

One diode half-wave rectifier



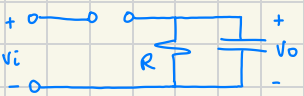


@ $t=0$, $V_o = 0$

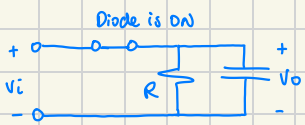
Assume: $RC \gg T$

During: $0 < t < T/4$: $V_o = V_i$

During: $t > T/4$: $V_o = V_p e^{-t/RC}$ [from circuit theory 1]



Capacitor is fully loaded [V_o would be temporarily greater than V_i since starts to drop]



V_r and ΔT need to be computed

$V_o = V_K$ @ $t = T - \Delta t$

we know that: $V_o(t) = V_p e^{\frac{-t}{RC}}$

$V_o(t-\Delta T) = V_K = V_p e^{-(T-\Delta T)/RC}$, Assume $\Delta T \ll T$

$\Rightarrow V_K \doteq V_p e^{-T/RC}$

we know that $RC \gg T$, hence we can use Taylor series approx: $e^x \doteq 1 + x$

In our case: $x = \frac{-T}{RC}$

$\Rightarrow V_K \doteq V_p \left(1 - \frac{T}{RC}\right)$

$V_p = V_r + V_K = V_r + V_p\left(1 - \frac{T}{RC}\right) = V_r + V_p - V_p\frac{T}{RC} \Rightarrow \cancel{V_p} = V_r + \cancel{V_p} - V_p\frac{T}{RC} \Rightarrow 0 = V_r - \frac{V_p T}{RC} \Rightarrow \boxed{V_r = \frac{V_p T}{RC}}$ $T = \frac{1}{f} \Rightarrow \boxed{V_r = \frac{V_p}{fRC}}$

Computing Δt

Since : $\Delta t \ll T$

we can say: $V_K = V_p \cos(\omega \Delta t)$

Δt is very small, hence use taylor series approx: $\cos(x) \approx 1 - \frac{x^2}{2}$

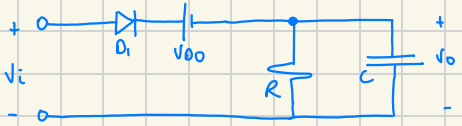
in our case: $x = \omega \Delta t$

$\rightarrow V_K = V_p \cos(\omega \Delta t) \approx V_p \left[1 - \frac{(\omega \Delta t)^2}{2}\right] \rightarrow V_K = V_p \left[1 - \frac{(\omega \Delta t)^2}{2}\right]$, $V_p = V_K + V_r \rightarrow \boxed{V_K = V_p - V_r}$

$\rightarrow V_p - V_r = V_p \left(1 - \frac{(\omega \Delta t)^2}{2}\right) = V_p - V_p \frac{(\omega \Delta t)^2}{2} \rightarrow V_p - V_r = V_p - V_p \frac{(\omega \Delta t)^2}{2} \rightarrow \not{V_p} - \not{V_p} = V_r - \frac{V_p (\omega \Delta t)^2}{2} \rightarrow V_r = \frac{V_p (\omega \Delta t)^2}{2}$

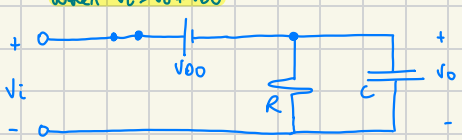
$\rightarrow (\omega \Delta t)^2 = \frac{2V_r}{V_p} \rightarrow \omega \Delta t = \sqrt{\frac{2V_r}{V_p}} \rightarrow \boxed{\Delta t = \frac{1}{\omega} \sqrt{\frac{2V_r}{V_p}}}$

Non-ideal diode : r_D : diode resistance can be ignored.



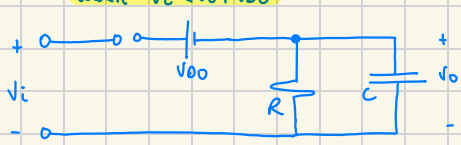
2 cases:

When $V_i > V_o + V_{DD}$

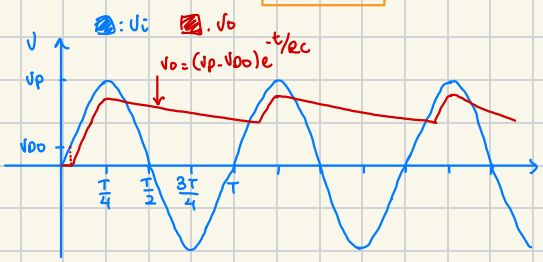


&

When $V_i < V_o + V_{DD}$



$V_{DD} = V_i - V_o \rightarrow V_{D_0} + V_o = V_i \rightarrow \boxed{V_o = V_i - V_{DD}}$



$\boxed{V_r = \frac{(V_p - V_{DD})}{RC} \frac{T}{}}$ $\quad$ $\boxed{V_r = \frac{(V_p - V_{DD})}{fRC}}$